



## The Development of a Podcast Motivations Scale for Taiwan

Suyu Chou, Wenchao Ma & Rebecca K. Britt

**To cite this article:** Suyu Chou, Wenchao Ma & Rebecca K. Britt (2023) The Development of a Podcast Motivations Scale for Taiwan, *Journal of Radio & Audio Media*, 30:2, 643-663, DOI: 10.1080/19376529.2022.2044818

**To link to this article:** <https://doi.org/10.1080/19376529.2022.2044818>



Published online: 28 Apr 2022.



Submit your article to this journal [↗](#)



Article views: 970



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 10 View citing articles [↗](#)



# The Development of a Podcast Motivations Scale for Taiwan

Suyu Chou , Wenchao Ma , and Rebecca K. Britt 

## ABSTRACT

The current study examined motivations for podcast listeners in Taiwan by constructing a preliminary podcast motivations scale. Theoretically grounded in uses and gratifications, the current research drew upon extant work in podcast motivations and proposed a new scale as we look to podcast engagement in a Taiwanese context, considering the global marketplace of podcast popularity. As podcasts have emerged as a popular form of entertainment in Taiwan, the study participants was composed of a Taiwanese sample, with the proposed podcast motivations scale tested through exploratory factor analysis (EFA), confirmatory factor analysis (CFA), and measurement invariance ( $N = 1,584$ ). Results elicited the 15-item podcast motivations scale. Three factors were identified and explained over fifty (54.1%) percent of the variance, consisting of *entertainment* (21.1%), *information-seeking* (18.4%), and *companionship* (14.6%). Overall, the results indicate that this new scale is a reliable measurement for understanding the motivations of podcast listeners in Taiwan, with implications for a global consumer base as we look to the future of podcast engagement.

## ARTICLE HISTORY

Received 17 November 2021

Revised 16 February 2022

Accepted 16 February 2022

Research shows that 57% of people in the US listened to podcasts in 2021, a number that is up from 55% in 2020 and 51% in 2019 (Edison Research and Triton Digital, 2021). Podcast listeners in the US have more than doubled compared to 2010 (Shoss, 2021), particularly as consumers tend to be highly engaged in multitasking behaviors, as the act of “listening” is often not the sole behavior associated with a podcast, as it requires selecting, downloading, and playing episodes while engaged with other activities. Such behaviors are largely reflective of the core tenets of uses and gratifications (Chan-Olmsted & Wang, 2020). As changes in the modern media landscape have evolved,

**CONTACT** Suyu Chou  [schou@crimson.ua.edu](mailto:schou@crimson.ua.edu)  Department of Journalism and Creative Media, University of Alabama. 484A Reese Phifer Hall, 901 University Blvd., Tuscaloosa, AL 35401, USA

© 2022 Broadcast Education Association

media multitasking behavior has shown that people often “switch” between media, and concurrently engage in both listening to podcasts while engaging in otherwise mundane tasks (Brasel & Gips, 2011).

Engaging in podcast entertainment—downloading, listening and playing episodes—has other advantages, as podcasts are, by and large, offered for free, and are versatile in their nature, offering information, entertainment, educational content, among other types of genres (McClung & Johnson, 2010). While podcasts have largely entered the mainstream media as a source of news and entertainment, the research on the motivations of podcast listeners is limited. Moreover, as podcasts have increasingly grown in popularity in countries outside of the US, including Taiwan (BiUh, 2021), research on the podcast has largely focused on a Western lens. Prior studies of Taiwanese users have dominantly focused on the high reach of social media platforms (e.g., Facebook; Alhabash, Chiang, & Huang, 2014), though podcast listeners have grown significantly since 2020 in Taiwan. According to the statistics of a local Taiwanese podcast hosting platform Firstory (BiUh, 2021), the number of downloads of podcasts in Taiwan in January 2021 was 579 times compared to January 2020 (INSIDE, 2021).

Given the growth of the podcast marketplace in Taiwan in a relatively short period of time, to better understand the growth and motivations of podcast audiences in Taiwan, this study sought to explore Taiwanese podcast listeners’ motivations and develop a comprehensive podcast motivations scale. Grounded in uses and gratifications, this study draws upon prior research, including podcast scale development (Chan-Olmsted & Wang, 2020; McClung & Johnson, 2010; Perks, Turner, & Tollison, 2019) and prior Taiwanese new media motivations studies (Chou, Lai, & Liu, 2013; Chou & Liu, 2016) which lends insight into the mobile usage of users and uses and gratifications as a theoretical framework.

## Literature Review

### *Podcasts as a Medium*

To provide an orientation of the history of podcasts and its trajectory in the new media landscape, we begin with a definition and history of podcasts. The word “podcast” is derived from a combination of “broadcasting” and “iPod,” and is defined as “a digital recording of a radio broadcast or similar program, made available on the internet for downloading to a personal audio player” (BBC News, 2005). Podcasts are audio and video files that can be downloaded to computers and portable media devices for “on the go” listening or playback later (McGarr, 2009). Audiences can subscribe to and automatically

download new episodes as they become available on their portable media device and they can listen to them whenever and wherever they want (McClung & Johnson, 2010).

The historical impact of podcasts is likewise significant: for instance, the word “podcast” was chosen as the word of the year in 2005 by the New Oxford American Dictionary (BBC News, 2005). In 2005, George W. Bush was the first president to have his weekly address delivered as a podcast. Fifteen years later in 2021, former President Barack Obama and rock star Bruce Springsteen launched their own podcast “Renegades: Born in the USA,” about masculinity, race, fatherhood, and the divisions that persist in American society (The New York Times, 2021). Podcasts have since moved beyond traditional broadcasting, such as radio. Ricky Gervais, who is one of the first big stars to create original video and audio material for the internet, noted that, “We don’t have a boss. We could do it when we wanted, for as long as we wanted. We could put it up there and then people would go to it at any time” (BBC News, 2006).

As such, the mainstream impact of podcasts is clear: the profound impact of major news corporations from *The New York Times* to *Fox News* to *CNN* have embraced podcasts and offered listeners free access to listen to their latest news (Casares, 2020). Shows such as *This American Life* is a weekly public radio show, listened to by 2.2 million people, and by another 2.5 million people from the weekly podcast (NPR, 2021). Recently, *This American Life* won the first Pulitzer Prize ever awarded to a radio show for audio reporting. The awarded episode, “The Out Crowd,” discussed issues related to a new Trump immigration policy. The award is a milestone not only for *This American Life*, but also for the entire medium, which has been energized over the last decade by the emergence of podcasting as a force in journalism and culture (The New York Times, 2020).

Podcasting, as a new form of audible media has taken time to blend into daily life—and to some extent—to potentially replace traditional broadcasting as a source of news and entertainment. Between 2009 and 2019, the number of Americans who had ever listened to a podcast more than doubled, growing from 22 to 51% (Statista, 2020). Recent research shows that 57% of people in the US listened to podcasts in 2021. This number is up from 55% in 2020 and 51% in 2019 (Edison Research and Triton Digital, 2021). In general, audiences of podcasts skew younger. More than half of Americans (56%) aged 12 to 34-year-olds listen to podcasts monthly. However, monthly podcast listening growing fastest among older Americans. From 2020 to 2021, the percentage of Americans ages 55 plus who listen to podcasts monthly increased by 18% (Edison Research and Triton Digital, 2021).

Nowadays, podcasts attract many audiences not only in America but all over the world. According to the data on the Podcastinsights website, currently there are over 2,000,000 active podcasts as of April 2021, including content in more than 100 languages (Podcastinsights, 2021).

### ***Podcasts in Taiwan***

While podcasts have been dominantly examined in a Western context, the growth of podcasts have likewise emerged globally, which brings opportunity for researchers and practitioners to leverage podcasts to disseminate information and campaigns for a global marketplace. In Taiwan, podcasts have emerged as a popular medium. In 2000, there were roughly 300 podcasts in Taiwan, though that number has only increased in the past 19 years (Udn, 2020), particularly in the past several years. For instance, in the first half of 2020, there were more than 870 new podcasts in the market (Chinatimes, 2020). According to Firstory, a podcast company in Taiwan, the number of downloads of Taiwan's podcasts in December 2020 was 446 times in January 2020; subsequently, the number of downloads in January 2021 increased by 579 times over the same period last year (INSIDE, 2021).

Notably, the podcast market in Taiwan may have experienced its growth in the past two years as a result of the ongoing COVID-19 pandemic, as individuals sought out new pastimes. In addition, celebrity and influencer figures postponed regular work during the pandemic and developed new podcasts during this time, which may have helped to spur additional audience growth during this time. Opinion leaders such as these who already have existing followers have, therefore, an increased likelihood of reaching new audiences via podcasts (Taipei Association of Advertising Agencies, 2020).

Podcasts in Taiwan have increased growth as content creators build new materials (SoundOn, 2020) and include celebrities, in addition to authoritative traditional media such as "CommonWealth," "udn Global News," and "The Real Story" (Yahoo News, 2021). Other programs, such as "Gooaye" serve as a general financial podcast, which is often a high-ranking podcast in Taiwan. The host, Xie Menggong, started operating his Facebook fan page in 2019 at the age of 26, and launched his podcast at the end of February 2020 (The News Lens, 2021). In that podcast, he shares general observations on the economy, international markets, and personal views on stock market investments in a relaxed and humorous tone. After starting his podcast, in just three months, his listeners increased to 200,000, quickly reaching the top three in the podcast rankings (Udn, 2020). Nowadays, according to the host of "Gooaye," Xie Menggong, the number of his listeners in 2021 is over 1,000,000.

According to the 2020 industry survey report of a local Taiwanese podcast company “SoundOn,” the top three Taiwanese users listening preferences included: *entertainment*, *interests*, and *professional knowledge*, with general topics including *society/culture*, *news/ politics*, and *entertainment/gossip* (SoundOn, 2020). “Bilingual News” covers international news, while “Taiwan No. 1 Commuter Brand” focuses on social phenomena and gossip (INSIDE, 2020). In addition to these genres of the podcast, there are increased types of podcasts in the media market, including sub-cultural, sexual, language learning, celebrity interviews, technology trends, self-improvement, and storytelling (SoundOn, 2020). The growth of podcasts in Taiwan is expected to continue, as those like the crime radio drama “Serial” allows a wider audience to engage with podcasts and enter the podcast market.

Studies of users in Taiwan and media engagement have typically examined the role of social media, including mobile messages (Chou et al., 2013; Chou & Liu, 2016) as well as theoretical undertakings examining preferences between platforms in television viewing and uses and gratifications (Chen, 2019). Additionally, this study referred to the 2020 Taiwanese podcast industry survey report (SoundOn, 2020) to establish survey items. Such an undertaking is useful as they consider a non-Western context. As such, we follow prior literature that has examined uses and gratifications as a theoretical framework to guide the current study to better understand users’ satisfaction, motivations, and context through which podcasts might be enjoyed by users (Chen, 2019; Chou & Liu, 2016).

### **Uses and Gratifications in the Context of Podcasts**

Wimmer and Dominick (2009) noted that in its earliest days, uses and gratifications (U&G) dominantly focused on the role of proactive involvement by audiences in the 1940s when given media choices. The active audience is the core tenet of U&G. Notably, podcast listeners tend to “align” with an active audience perspective because, by their very nature, *must* actively search for a podcast, subscribe, and click the program to play when they have time. Consequently, U&G is an appropriate approach to explore the reasons why audiences listen to podcasts (Katz, Blumler, & Gurevitch, 1973; Katz, Haas, & Gurevitch, 1973).

The extant research on podcast audiences has examined different topics, ranging from exploring factors of new media adoption (Li & Zeng, 2011) to audience engagement cultivation (Wrather, 2016). However, podcast-related studies have primarily examined how they can be utilized as a teaching tool for learning (Evans, 2008; McGarr, 2009; Şendağ, Gedik, & Toker, 2018; Yeh,

2017). Little research has examined the motivations of podcast listeners, though it would be particularly useful to understand what motivates and gratifies audiences when they engage with such media.

Traditionally, as Mendelsohn (1964) noted, the motivations of listening to radio resulted in the feeling of companionship, altering one's mood, relieving boredom, obtaining information, participating in events vicariously, and assisting with social interactions. Such gratifications are crucial to understand; as Hancock and McMurtry (2018) note, what ultimately distinguishes podcasts from the radio is its accessibility, portability, user control options, and its lack of censorship systems that radio has, which makes it a more freeform option. Consequently, considering features of podcast motivations, McClung and Johnson (2010) explored why individuals use podcasts by analyzing 354 active podcast users. Their results identified five reasons: entertainment, time-shifting, library building, advertising, and social aspects, which explain nearly sixty (58.3%) percent of the variance. Importantly, Perks et al. (2019) relied on previous qualitative findings (Perks & Turner, 2019) and designed a two-study survey to develop a podcast U&G scale. Their factor analysis identified four factors: edutainment, storytelling transportation, social engagement, and multitasking.

Boling and Hull (2018) explored the true-crime podcast audience by using an online survey ( $n = 308$ ). This study found that three motivations were prominent for users: entertainment, convenience, and boredom (Boling & Hull, 2018). Recently, Chan-Olmsted and Wang (2020) based on previous podcast motivations studies and industry reports (Chan, 2014; Haridakis & Hanson, 2009; Khan, 2017; Nabi, Stitt, Halford, & Finnerty, 2006; Papacharissi & Mendelson, 2007; Rubin, 1983; Ruggiero, 2000; Sundar & Limperos, 2013), and analyzed 2,000 regular podcast users. The results indicated that entertainment, information, and audio platform superiority were the most important motivations for podcast consumption (Chan-Olmsted & Wang, 2020).

Podcasting is still a new medium for many countries such as Taiwan. Little academic attention has been paid to podcasting in Taiwanese and global contexts; as such, this study offers insight into Taiwan audiences to help build a research agenda for global podcast research that can aid understanding of the motivations and gratifications of audiences. Likewise, there is little research on understanding the podcast audiences in Taiwan, in particular with regards to their listening motivations and usage patterns. Given the growth of podcasting in 2020, this study seeks to fill this gap and explore Taiwanese podcast users' podcasting motivations.

## Method

### Measures

Perks et al. (2019) developed an 18-item podcast U&G scale. In the current study, a total of 10 items are adapted from Perks et al. (2019). Other items come from previous studies and have been revised for this study (Casares, 2020; Chan, 2014; Chan-Olmsted & Wang, 2020; McClung & Johnson, 2010). To develop the items in the scale, the authors based the items generation process on the literature review.

Specifically, there were several steps that influenced how we choose the items. First, the authors referred to the scale from Perks et al. (2019). However, only three dimensions were included in this study: “controlling edutainment,” “storytelling transportation” and “social/multitasking.” In the present study, based on the literature review, the authors in the current study believe entertainment and education are two different concepts to conduct further research separately instead of combining them into one factor, as Perks et al did with “controlling edutainment.” Moreover, because we developed this as a preliminary scale in a Taiwanese context, the authors embraced other possible motivations from previous validated studies such as “companionship,” and “identification/social interaction” (Chan-Olmsted & Wang, 2020).

Furthermore, to revise and redesign the scale to evaluate podcast listeners in Taiwan appropriately, the authors also referred to the Taiwanese podcast industry reports and the discussions on popular podcast Facebook fan pages. For example, according to the 2020 industry survey report of a local Taiwanese podcast company “SoundOn,” the top three Taiwan podcasts’ listening purposes are: entertainment, interests, and professional knowledge (SoundOn, 2020). Eventually, there were a total of five major dimensions that reflected our conceptualization of podcast listening motivations. These five dimensions are: “information-seeking,” “entertainment,” “companionship,” “identification/social interaction,” and “convenience/multi-tasking.”

The first factor, “information-seeking” includes statements about how podcasts make people educated and well-informed. The second factor, “entertainment” includes statements about how podcasts make people feel relaxed and are viewed as fun. The third factor, “companionship” includes statements about how podcasts make people less lonely and how listeners view their podcast hosts as friends. The fourth factor, “identification/social interaction” includes statements about how podcasts improve listeners’ understanding about who they are and how podcasts could reaffirm their personal values. The fifth factor, “convenience/multi-tasking” includes statements about easily accessing podcasts when listeners want and the ability to



multi-task while listening to podcasts. Eventually, this study designed the preliminary podcast motivations items with 30 items covering the above five motivations (see Appendix 1). Each of the items will be measured on a 5-point Likert-type scale (1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree). The authors translated the English questionnaire to the Taiwanese version and tested the survey by four Taiwanese podcast listeners and amended its delivery accordingly to make it easily understood by participants. This study was reviewed and approved by the Research Review Committee at The University of Alabama, April 16, 2021 (NO. 21,034,482).

### ***Survey and Recruitment***

Study participants were recruited from podcast hosts' social media websites. These podcast hosts were picked according to the rank of Apple podcast in Taiwan and the most popular programs in 2020 which was surveyed by Taiwanese podcasts string platform KKBOX (TechNews, 2021). The authors invited a total of 15 podcast hosts the survey invitation. They were given an e-mail consisting of a brief introduction of the current study and a request for participation. Podcast hosts were asked to distribute the online survey. Four podcast hosts replied and were willing to help distribute the online survey on their Facebook fan page or Instagram. The genres of these four podcasts were diversified, including society and culture, news and politics, and entertainment and gossip. Therefore, participants would be Taiwanese or people who are familiar with Mandarin and follow the Taiwanese podcasters' social media. Additionally, one filter question in the survey is, "Where do you live in?" (The choices are: northern part of Taiwan, southern part of Taiwan, eastern part of Taiwan, western part of Taiwan, and Taiwanese living abroad) to make sure participants are Taiwanese. The survey was distributed online through Google forms from May 29<sup>th</sup> to June 21<sup>th</sup> 2021.

Upon accessing the survey, participants were given the informed consent materials. Upon agreement, participants were presented with the 30 preliminary podcast U&G scale items. They were also asked to provide basic demographic and relevant podcast listenership information. There were 43 questions in the survey. As incentives, the researchers provided a total of 10 vouchers (7–11 convenient store vouchers, each worth about 18 dollars) for a drawing for a prize. The questionnaire in this study was anonymous. Every participant who was willing to leave their e-mail address and nicknames had an equivalent chance to win a voucher as a prize.

## Results

Based on the questionnaire collected from podcast hosts, a total of 1,584 podcast listeners answered the online survey. Of the participants, 45.8% are male, 53.8% are female.

According to the population census from the Ministry of the Interior in Taiwan, 49.6% are male, 50.4% are female (ETtoday News, 2020). Therefore, the participants were skewed toward being female, which is aligned with the result of the previous industry report that the ratio of male to female podcast audiences in 2020 is about 41:59. In general, the number of female podcast listeners is higher than the number of males in Taiwan. Regarding education level, participants tend to be highly educated, 90% report having completed college or higher professional program, that is higher than the average education level (47% of people have a college degree or higher) in the latest statistic in Taiwan (Ministry of the Interior, 2021). To perform exploratory and confirmatory analyses, we split the sample into two randomized halves.

Given that factor analysis usually requires at least 10 participants per variable to achieve replicable findings (Norman & Streiner, 2008), and two to three times amount to account for all the parameters being estimated (Wang & Wang, 2019). The sample size ( $n_1 = 791$ ,  $n_2 = 793$ ) in this study met the above requirements and beyond the recommended sample size. Before we proceeded with the EFA, we checked the KMO statistic and Bartlett's test. The KMO test of sampling adequacy was .97, which, according to Kaiser (1974), indicated that the data were meritorious. The Bartlett's test is statistically significant ( $p < .001$ ), suggesting that the data were factorable (Williams, Onsman, & Brown, 2010).

### *Exploratory Factor Analysis*

An exploratory factor analysis (EFA) was conducted to understand the factor structure of podcast motivations in Taiwan ( $n_1 = 791$ , 45% male and 55% female). It was performed on 30 items of podcast motivations scale for the participants. The dimensionality of the test was determined using parallel analysis (Timmerman & Lorenzo-Seva, 2011), which suggests three factors: "Entertainment (21.1%)," "Information-seeking (18.4%)" and "Companionship (14.6%)." In total, these three factors explain over fifty (54.1%) percent of the variance. The EFA with the oblique rotation using the promax method was performed by setting the number of dimensions at three. Three items were dropped due to cross-factor loading, and the other three items were dropped because of the unsatisfying loading size, leaving 24 questions for the next analysis.

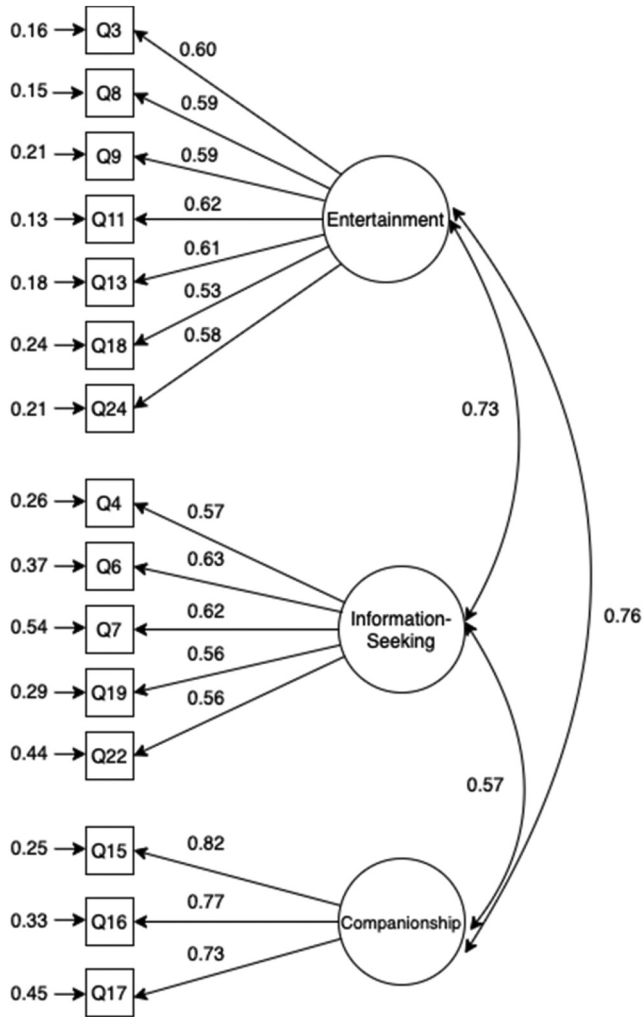
### **Confirmatory Factor Analysis**

To assess the robustness of the identified structure, the exploratory model was assessed for its structural validity by using Confirmatory Factor Analysis (CFA) (see [Figure 1](#)). Using the second half of the data ( $n_2 = 793$ , 47% male and 53% female), the remaining items were analyzed using CFA to evaluate whether the three-factor, 24 item scale proposed above would fit data adequately. According to MacCallum, Browne, and Sugawara (1996), Hu and Bentler (1999) as well as Kenny and McCoach (2003), the most commonly used criteria for a good fit are: Comparative Fit Index (CFI), Tucker–Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean-square Residual (SRMR). The criterion of  $CFI \geq .90$ ,  $TLI \geq .90$ ,  $RMSEA \leq .08$ , and  $SRMR \leq .08$  are seen as a good standard of model fit (Hu & Bentler, 1999; Kline, 2015; Maydeu-Olivares, Shi, & Rosseel, 2018; Taasobshirazi & Wang, 2016; Van de Schoot, Lugtig, & Hox, 2012).

Based on the aforementioned criteria, this initial model seems not fit data adequately ( $X^2 = 12,186.308$  ( $df = 253$ ;  $p < .001$ ),  $CFI = .879$ ;  $TLI = .865$ ;  $RMSEA = .089$ ;  $RMSEA$  90% CI [.085, .093];  $SRMR = .085$ ). By carefully examining the modification index, which suggests where the misfit occurs, we eliminated 9 items (see [Table 1](#)) and performed a CFA with the remaining 15 items. The three-factor model with 15 items exhibited a good model-data fit with  $X^2 = 7404.29$  ( $df = 105$ ;  $p < .001$ ),  $CFI = .955$ ;  $TLI = .945$ ;  $RMSEA = .069$ ;  $RMSEA$  90% CI [.063, .076];  $SRMR = .046$ .

### **Measurement Invariance**

Measurement invariance assesses the psychometric equivalence of a construct across groups or time to demonstrate that a construct has the same meaning to those groups or across repeated measurements (Putnick & Bornstein, 2016). The function of the measurement invariance is to define conditions that should be fulfilled before inferences especially, in cross-national/cultural studies. In other words, measurement invariance suggests that using the same questionnaire in different situations, such as countries, times, or conditions, does measure the same construct in the same way (Chen, 2008; Davidov, Schmidt, & Schwartz, 2008; Millsap, 2012). Most tests of measurement invariance involve configural, metric, scalar, and strict models. The four measurement invariance steps considered are: (1) configural model, equivalence of model form; (2) metric model (weak factorial), equivalence of factor loadings; and (3) scalar model (strong factorial), equivalence of item intercepts or thresholds, and (4) strict model, also known as residual or invariant uniqueness (Putnick & Bornstein, 2016). To compare two models, Putnick and Bornstein (2016) stated that for sample



**Figure 1** CFA Model of Entertainment, Information-seeking, and Companionship

sizes with adequate power, equal group sizes, and mixed invariance, a criterion of a  $-.01$  change in CFI for nested models is commonly used (Cheung & Rensvold, 2002), paired with changes in RMSEA of .015 and SRMR of .030 (for metric invariance) or .015 (for scalar or strict invariance).

In this study, we examined the measurement invariance of the developed motivations scale across the gender groups. The configural model (M1) was first established and it can be observed from Table 2 that M1 had a good fit to the data. Secondly, the configural model was compared to the metric model (M2) that constrained factor loadings equal across gender groups. Chi-square difference tests indicated no significant change in model fit between

**Table 1** Podcast Motivations Scale Items

| Scale Items                                                                                             | Loadings |
|---------------------------------------------------------------------------------------------------------|----------|
| <b>Factor 1: Entertainment; 7 items (<math>\alpha = .93</math>)</b>                                     |          |
| 3) I like to learn about other people's experiences through listening to podcasts                       | 0.60     |
| 8) Listening to podcasts is my entertainment                                                            | 0.59     |
| 9) Listening to podcasts is fun                                                                         | 0.59     |
| 11) Listening to podcasts is relaxing                                                                   | 0.62     |
| 13) Listening to podcasts makes me happy                                                                | 0.61     |
| 18) I like my hosts to share their personal stuff during the show                                       | 0.53     |
| 24) I have searched my favorite podcasts' information online or followed their social media             | 0.58     |
| <i>*14) Listening to podcasts helps me pass the time</i>                                                |          |
| <i>*28) I like to listen to podcasts because I can do other things while listening to them</i>          |          |
| <i>*29) When I am doing other things while listening to podcasts, the time goes by faster</i>           |          |
| <b>Factor 2: Information-seeking; 5 items (<math>\alpha = .82</math>)</b>                               |          |
| 4) I can enrich myself when I listen to podcasts                                                        | 0.57     |
| 6) I think there is abundant information I can get when I listen to podcasts                            | 0.63     |
| 7) Compared to other media, listening to podcasts can help me understand a certain topic                | 0.62     |
| 19) Listening to podcasts improves my understanding about what kind of person I am                      | 0.56     |
| 22) Listening to podcasts inspires my ideas or thoughts                                                 | 0.56     |
| <i>*26) Listening to podcasts is a convenient way to get information</i>                                |          |
| <i>*27) Listening to podcasts is easier to get information compared to other forms of media</i>         |          |
| <i>*30) Listening to podcasts helps me make good use of the fragmented or in-between time</i>           |          |
| <b>Factor 3: Companionship; 3 items (<math>\alpha = .84</math>)</b>                                     |          |
| 15) Listening to podcasts makes me feel accompanied                                                     | 0.82     |
| 16) Listening to podcasts makes me feel less lonely                                                     | 0.77     |
| 17) Listening to podcasts makes me feel the host is one of my friends                                   | 0.73     |
| <i>*10) Listening to podcasts helps me relieve stress</i>                                               |          |
| <i>*20) Listening to podcasts helps me know more about myself</i>                                       |          |
| <i>*25) I am interested in the discussions and interactions on my favorite podcasters' social media</i> |          |

\*Questions were dropped out after conducting CFA.

configural and metric models in gender difference (see Table 2). The metric model had a good fit to the data and the changes in fit index indicated no meaningful decrement in fit from configural to the metric model for gender groups ( $\Delta\text{CFI} = .000$ ,  $\Delta\text{RMSEA} = -.002$ ,  $\Delta\text{SRMR} = .002$ ).

Furthermore, we compared the metric model (M2) to the scalar model (M3) that constrained factor loadings and thresholds across the gender groups. When item intercepts were made invariant by gender, chi-square difference tests indicated significant change ( $p < .001$ ) in model fit between metric and scalar models across gender groups.

However, differences in practical fit were minimum, less than .01 for the CFI, the most severe criteria ( $\Delta\text{CFI} = -.002$ ,  $\Delta\text{RMSEA} = .000$ ,  $\Delta\text{SRMR} = .002$ ). Likewise, the same happened when strict model (M4) was tested, as practical differences in fit were irrelevant ( $\Delta\text{CFI} = -.006$ ,  $\Delta\text{RMSEA} = .002$ ,  $\Delta\text{SRMR} = .003$ ). Overall, we decided to accept our results as evidence for measurement invariance as changes in CFI, RMSEA, and SRMR across the increasingly constrained models did not indicate any meaningful decrement in model fit (Chen, 2007; Cheung & Rensvold, 2002). Accordingly, the set of equivalences are considered tenable, and the podcast motivations' factors could be considered equivalent by gender.

**Table 2** Goodness-of-Fit Indicators and Comparison of Measurement Models across Gender

| Model                | $\chi^2$ <i>df</i> | CFI  | RMSEA<br>(90% CI) | SRMR | Model<br>comp | $\Delta\chi^2$ | $\Delta$ CFI | $\Delta$ RMSEA | $\Delta$ SRMR | Decision |
|----------------------|--------------------|------|-------------------|------|---------------|----------------|--------------|----------------|---------------|----------|
| M1: Configural model | 862.91*** (174)    | .957 | .065(.072-.059)   | .043 | —             | —              | —            | —              | —             | —        |
| M2: Metric model     | 872.97*** (186)    | .958 | .063(.069-.057)   | .045 | M1            | 9.086(12)      | 0.000        | −0.002         | 0.002         | Accept   |
| M3: Scalar model     | 919.03*** (198)    | .955 | .063(.069-.057)   | .047 | M2            | 46.607*** (12) | −0.002       | 0.000          | 0.002         | Accept   |
| M4: Strict           | 1042.81(213)       | .949 | .065(.070-.059)   | .049 | M3            | 47.126*** (15) | −0.006       | 0.002          | 0.003         | Accept   |

Note. N = 1,579; Male = 726; Female = 853.  
\*\*\* $p \leq .001$ .

## Discussion

The purpose of this study was to develop a podcast motivations scale in a Taiwanese context. The eventual results identified three-factor, 15 items podcast motivations scale, including “Entertainment (7 items,  $\alpha = .93$ ),” “Information-seeking (5 items,  $\alpha = .82$ )” and “Companionship (3 items,  $\alpha = .84$ ).” Among all the motivations of listening to podcasts, “Entertainment (21.1%)” is the strongest factor followed by “Information-seeking (18.4%),” and “Companionship (14.6%),” totally explaining over fifty (54.1%) percent of the variance.

Compared to the initial five factors hypothesis, the two factors “Identification/Social interaction,” “Convenience/Multi-tasking,” were not significant enough to construct an independent factor. To evaluate the factor “Identification/Social interaction,” originally, seven items were generated (see Appendix 1). Through the EFA, four items were dropped, and two items were combined to the factor “Information-seeking”: (19) Listening to podcasts improves my understanding about what kind of person I am, and (22) Listening to podcasts inspires my ideas or thoughts (see Table 1). Likewise, five items were generated to evaluate the factor “Convenience/Multi-tasking (See Appendix 1).” After the EFA three items were combined to the factor “Information-seeking”: (26) Listening to podcasts is a convenient way to get information, (27) Listening to podcasts is easier to get information compared to other forms of media, and (30) Listening to podcasts helps me make good use of the fragmented or in-between time. Another two items were combined to the factor “Entertainment”: (28) I like to listen to podcasts because I can do other things while listening to it, and (29) When I am doing other things while listening to podcasts, the time goes by faster. To make the scale reach the fit indices, all the five items under the “Convenience/Multi-tasking” were deleted in conducting the CFA, which made the CFI increase from .879 to .882. Given that we looked at podcast motivations as an active audience approach, however, these findings make sense. As U&G research suggests—and as active audience approaches suggest in podcast motivations research—users are engaging with podcasts as a new medium for *entertainment* and *information*, as well as *companionship* when going about their daily lives, rather than other factors.

Although the items related to the factors “Identification/Social interaction” and “Convenience/Multi-tasking” were deleted or adjusted to other factors such as “Information-seeking,” and “Entertainment” through EFA, we can explain that the convenience/multi-tasking factor is included in the primary motivations for Taiwanese podcasters listeners. Namely, people are seeking entertainment and knowledge as their fundamental motivations for tuning in to podcasts. Listening to podcasts conveniently is their secondary requirement. The alternative explanation for this phenomenon is that the

convenience of media usage might be naturally rooted in the modern world, therefore, the factor convenience/multi-tasking is not showing obvious differences. Nevertheless, the concept of multi-tasking has been widely examined among American podcast studies. On a more global level, there is the potential for further studies to examine the role of multitasking and podcast usages, such as examining the role of high-achievers and the podcasts they listen to; how do podcasters discuss their creations online, and ultimately yield insights into media behaviors.

Additionally, compared to the previous podcast studies conducted in the context of the Western culture, the basic motivations for listening to podcasts were similar, yet offer some distinct findings that are important moving forward in developing an agenda for Taiwan and a more global agenda for research in podcast motivations scholarship. All told, prior podcast motivation studies all included the same factor—*entertainment*. Perks et al. (2019) had four factors: *controlling edutainment*, *storytelling transportation*, *social engagement*, and *multitasking*. McClung and Johnson (2010) had four factors: *entertainment*, *timeshifting*, *library building*, a favorable view of *advertising*, and remarked on a social aspect of the podcasts that centers around discussing the shows with other fans. Chan-Olmsted and Wang (2020) had three factors: *entertainment*, *information*, and *audio platform superiority*. The current study had three main motivations: *entertainment*, *information-seeking*, and *companionship*. Notably, the factor companionship did not appear in prior research, suggesting that it may be an important variable to examine moving forward, especially given that motivations for further engagement with audio-visual media may mean that individuals are increasingly seeking companionship in some fashion may do so from the information and entertainment they consume. The companionship items in this study were generated by referring to Chan-Olmsted and Wang's study (2020), however, in their study, the factor was not significant. The difference might be caused by cultural differences. More studies should be done to examine this presumption. As such, the unique contribution of this study is that it provides a scale of podcast motivations for users in Taiwan, with directions for research that can be applied in future contexts of audience engagement.

## Limitations and Future Research Directions

Limitations in this study include sampling and credibility. First, participants in this study were a convenience sample rather than a random sample. They were recruited from popular podcast hosts' social media websites. While the convenience sample used in this study is not ideal, it served a practical



purpose for the results of the scale development. Hence, the errors due to the convenience sample in this study were inevitable and may have caused potential biases in the results.

Additionally, although the current podcast motivations scale was based on uses and gratifications and revised from a tested scale (Perks et al., 2019), the credibility of the scale is still worth examining. Notably, after dropping nine items in CFA, no extra data was available to re-test the eventual model. As such, the current study serves as the first step in a model for developing a podcast motivations scale. Additionally, to further access the validity of the scale, future studies should include listeners' usage behaviors and evaluate the relationship between those variables accordingly to further confirm the validity.

As whole, this study integrated U&G and prior scales using podcasts and empirical findings into a new scale that can be used on the role of podcast motivations for Taiwanese users. This research makes important contributions to developing the podcast motivations scale and understanding podcast audiences in Taiwan. The current study's research participants are from Taiwan, where the podcast industry has experienced exponential growth since 2020. Therefore, this study can be a reference for countries where podcasts are still a new form of media. Moreover, although this research served as a preliminary study for developing a podcast motivations scale, the results shed the light on the possibility of generating profits for podcasters. For example, the results showed that companionship is one crucial factor; therefore, if the host were the audiences' friend, it might be easier for advertisements to promote productions on podcasts. Future studies could also find the link between podcasts audiences' motivation and marketing.

## Disclosure Statement

No potential conflict of interest was reported by the author(s).

## Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

## Notes on contributors

*Suyu Chou* is a graduate student at the College of Communication & Information Sciences, University of Alabama. Her research interests lie in the intersection of public anxiety on social media and crisis management.

**Wenchao Ma** is an assistant professor at the Department of Educational Studies in Psychology, Research Methodology & Counseling, University of Alabama. His research interests lie in psychometrics and its applications in education, psychology, and other social sciences.

**Rebecca K. Britt** is an associate professor in the College of Communication & Information Sciences, University of Alabama. Her research centers on the role of health crises, social systems and the role of behaviors and networks; online collectives and computational social science.

## ORCID

Suyu Chou  <http://orcid.org/0000-0002-8999-9302>

Wenchao Ma  <http://orcid.org/0000-0002-6763-0707>

Rebecca K. Britt  <http://orcid.org/0000-0002-4595-4604>

## References

- Alhabash, S., Chiang, Y., & Huang, K. (2014). MAM & U&G in Taiwan: Differences in the uses and gratifications of Facebook as a function of motivational reactivity. *Computers in Human Behavior*, 35, 423–430. doi:10.1016/j.chb.2014.03.033
- BBC News (2005). Wordsmiths hail podcast success. Retrieved from <https://bbc.in/3kOsM5>
- BBC News (2006). Future of TV: Ricky Gervais. Retrieved from <http://news.bbc.co.uk/2/hi/entertainment/6164812.stm>
- BiUh, W. (2021, April 16) *Look at the \$1 billion business opportunity of the sound economy from the Podcast!* Udn. <https://bit.ly/3poRH48>
- Boling, K. S., & Hull, K. (2018). Undisclosed information—Serial is My Favorite Murder: Examining motivations in the true crime podcast audience. *Journal of Radio & Audio Media*, 25(1), 92–108. doi:10.1080/19376529.2017.1370714
- Brasel, S. A., & Gips, J. (2011). Media multitasking behavior: Concurrent television and computer usage. *Cyberpsychology, Behavior, and Social Networking*, 14(9), 9. doi:10.1089/cyber.2010.0350
- Casares, D. R., Jr. (2020). Embracing the Podcast Era: Trends, opportunities, & implications for counselors. *Journal of Creativity in Mental Health*, 1–16. doi:10.1080/15401383.2020.1816865
- Chan, M. (2014). Social identity gratifications of social network sites and their impact on collective action participation. *Asian Journal of Social Psychology*, 17(3), 229–235. doi:10.1111/ajsp.12068
- Chan-Olmsted, S., & Wang, R. (2020). Understanding podcast users: Consumption motives and behaviors. *New Media & Society*. doi:10.1177/1461444820963776
- Chen, F. F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. *Structural Equation Modeling: A Multidisciplinary Journal*, 14(3), 464–504. doi:10.1080/10705510701301834
- Chen, F. F. (2008). What happens if we compare chopsticks with forks? The impact of making inappropriate comparisons in cross-cultural research. *Journal of Personality and Social Psychology*, 95(5), 1005. doi:10.1037/a0013193

- Chen, Y. N. K. (2019). Competitions between OTT TV platforms and traditional television in Taiwan: A niche analysis. *Telecommunications Policy*, 43(9). doi:10.1016/j.telpol.2018.10.006
- Cheung, G. W., & Rensvold, R. B. (2002). Evaluating goodness-of-fit indexes for testing measurement invariance. *Structural Equation Modeling*, 9(2), 233–255. doi:10.1207/S15328007SEM0902\_5
- Chinatimes. (2020). Podcasts in Taiwan: The ear economy is coming. Retrieved from <https://bit.ly/3FwG8gC>
- Chou, W. H., Lai, Y. T., & Liu, K. H. (2013). User requirements of social media for the elderly: A case study in Taiwan. *Behavior & Information Technology*, 31(9), 920–937. doi:10.1080/0144929X.2012.681068
- Chou, M., & Liu, C. H. (2016). Mobile instant messengers and middle-aged and elderly adults in Taiwan: Uses and gratifications. *International Journal of Human-Computer Interaction*, 31(11), 835–846. doi:10.1080/10447318.2016.1201892
- Davidov, E., Schmidt, P., & Schwartz, S. H. (2008). Bringing values back in: The adequacy of the European Social Survey to measure values in 20 countries. *Public Opinion Quarterly*, 72(3), 420–445. doi:10.1093/poq/nfn035
- Edison Research and Triton Digital. (2021). Podcast Statistics for 2021 – Charts and Data. Retrieved from <https://bit.ly/3sGrCPE>
- ETtoday News. (2020). The latest survey: “There are more women than men.” It’s hardest for a woman to find a husband. Retrieved from <https://bit.ly/32KdAS2>
- Evans, C. (2008). The effectiveness of m-learning in the form of podcast revision lectures in higher education. *Computers & Education*, 50(2), 491–498. doi:10.1016/j.compedu.2007.09.016
- Hancock, D., & McMurtry, L. (2018). ‘I know what a podcast is’: Post-serial fiction and podcast media identity. In Llinares, D., Fox, N. (eds) *Podcasting* (pp. 81–105). Cham: Palgrave Macmillan.
- Haridakis, P., & Hanson, G. (2009). Social interaction and co-viewing with YouTube: Blending mass communication reception and social connection. *Journal of Broadcasting & Electronic Media*, 53(2), 317–335. doi:10.1080/08838150902908270
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, 6(1), 1–55. doi:10.1080/10705519909540118
- INSIDE. (2020). Taiwan Podcast Data: The first market survey data, decrypting the gold content of the Podcast! Retrieved from <https://bit.ly/3JlQrXa>
- INSIDE. (2021). 2020 Taiwan podcast industry report for the second half of the year. Retrieved from <https://bit.ly/3H4DCyu>
- Kaiser, H. F. (1974). An index of factorial simplicity. *Psychometrika*, 39(1), 31–36. doi:10.1007/BF02291575
- Katz, E., Blumler, J. G., & Gurevitch, M. (1973). Uses and gratifications research. *The Public Opinion Quarterly*, 37(4), 509–523. doi:10.1086/268109
- Katz, E., Haas, H., & Gurevitch, M. (1973). On the use of the mass media for important things. *American Sociological Review*, 38(2), 164–181. doi:10.2307/2094393
- Kenny, D. A., & McCoach, D. B. (2003). Effect of the number of variables on measures of fit in structural equation modeling. *Structural Equation Modeling*, 10(3), 333–351. doi:10.1207/S15328007SEM1003\_1
- Khan, M. L. (2017). Social media engagement: What motivates user participation and consumption on YouTube? *Computers in Human Behavior*, 66, 236–247. doi:10.1016/j.chb.2016.09.024

- Kline, R. B. (2015). *Principles and practice of structural equation modeling*. Guilford publications.
- Li, X., & Zeng, L. (2011). Technology attributes, perceived value of information, and social utility: Predicting podcast adoption and use. *Southwestern Mass Communication Journal*, 27, 1.
- MacCallum, R. C., Browne, M. W., & Sugawara, H. M. (1996). Power analysis and determination of sample size for covariance structure modeling. *Psychological Methods*, 1(2), 130. doi:10.1037/1082-989X.1.2.130
- Maydeu-Olivares, A., Shi, D., & Rosseel, Y. (2018). Assessing fit in structural equation models: A Monte-Carlo evaluation of RMSEA versus SRMR confidence intervals and tests of close fit. *Structural Equation Modeling: A Multidisciplinary Journal*, 25(3), 389–402. doi:10.1080/10705511.2017.1389611
- McClung, S., & Johnson, K. (2010). Examining the motives of podcast users. *Journal of Radio & Audio Media*, 17(1), 82–95. doi:10.1080/19376521003719391
- McGarr, O. (2009). A review of podcasting in higher education: Its influence on the traditional lecture. *Australian Journal of Educational Technology*, 25(3), 309–321.
- Mendelsohn, H. (1964). Broadcast vs. personal sources of information in emergent public crises: The Presidential assassination. *Journal of Broadcasting & Electronic Media*, 8(2), 147–156. doi:10.1080/08838156409386098
- Millsap, R. E. (2012). *Statistical approaches to measurement invariance*. Routledge.
- Ministry of the Interior, R.O.C, Taiwan. (2021). Taiwan population 2021. Retrieved from <https://bit.ly/3qAleXr>
- Nabi, R. L., Stitt, C. R., Halford, J., & Finnerty, K. L. (2006). Emotional and cognitive predictors of the enjoyment of reality-based and fictional television programming: An elaboration of the uses and gratifications perspective. *Media Psychology*, 8(4), 421–447. doi:10.1207/s1532785xmep0804\_5
- Norman, G. R., & Streiner, D. L. (2008). *Biostatistics: The bare essentials*. PMPH USA (BC Decker).
- NPR. (2021). Podcast directory. The introduction of “This American Life.” Retrieved from <https://n.pr/3ps3ePD>
- Papacharissi, Z., & Mendelson, A. L. (2007). An exploratory study of reality appeal: Uses and gratifications of reality TV shows. *Journal of Broadcasting & Electronic Media*, 51(2), 355–370. doi:10.1080/08838150701307152
- Perks, L. G., & Turner, J. S. (2019). Podcasts and productivity: A qualitative uses and gratifications study. *Mass Communication and Society*, 22(1), 96–116. doi:10.1080/15205436.2018.1490434
- Perks, L. G., Turner, J. S., & Tollison, A. C. (2019). Podcast uses and gratifications scale development. *Journal of Broadcasting & Electronic Media*, 63(4), 617–634. doi:10.1080/08838151.2019.1688817
- Podcastinsights. (2021). Podcast stats & facts. Retrieved from <http://bit.ly/3c00yRE>
- Putnick, D. L., & Bornstein, M. H. (2016). Measurement invariance conventions and reporting: The state of the art and future directions for psychological research. *Developmental Review*, 41, 71–90. doi:10.1016/j.dr.2016.06.004
- Rubin, A. M. (1983). Television uses and gratifications: The interactions of viewing patterns and motivations. *Journal of Broadcasting*, 27(1), 37–51. doi:10.1080/08838158309386471
- Ruggiero, T. E. (2000). Uses and gratifications theory in the 21st century. *Mass Communication & Society*, 3(1), 3–37. doi:10.1207/S15327825MCS0301\_02

- Şendağ, S., Gedik, N., & Toker, S. (2018). Impact of repetitive listening, listening-aid and podcast length on EFL podcast listening. *Computers & Education*, 125, 273–283. doi:10.1016/j.compedu.2018.06.019
- Shoss, D. (2021). Tech talk – All things podcast. Retrieved from <http://bit.ly/2Por2Ez>
- SoundOn. (2020). H1 podcast industrial survey report. Retrieved from <https://docsend.com/view/7cegnvyirp9j65z>
- Statista. (2020). Share of US population that have listened to an audio podcast in the last month from 2008 to 2020. Retrieved from <https://www.statista.com/statistics/270365/audio-podcast-consumption-in-the-us/>
- Sundar, S. S., & Limperos, A. M. (2013). Uses and grats 2.0: New gratifications for new media. *Journal of Broadcasting & Electronic Media*, 57(4), 504–525. doi:10.1080/08838151.2013.845827
- The New York Times. (2020). ‘This American Life,’ Now a Pulitzer Winner, Is Once More a Pioneer. Retrieved from <https://www.nytimes.com/2020/05/07/arts/this-american-life-pulitzer.html>
- The New York Times. (2021). Barack Obama and Bruce Springsteen: The Latest Podcast Duo. Retrieved from <http://nyti.ms/3c0He6K>
- The News Lens. (2021). Interview: “Stock Cancer” Xie Menggong. Retrieved from <https://bit.ly/3Jmldzd>
- Taasobshirazi, G., & Wang, S. (2016). The performance of the SRMR, RMSEA, CFI, and TLI: An examination of sample size, path size, and degrees of freedom. *Journal of Applied Quantitative Methods*, 11(3), 31–39.
- Taipei Association of Advertising Agencies. (2020). Taiwan’s first year of podcast: Ear economy grows against the trend. Retrieved from <https://bit.ly/3mDCVnN>
- TechNews (2021). KKBOX: The most popular podcasts: Bilingual News, Stock Cancer, Taiwan No. 1 commuter brand. Retrieved from <https://bit.ly/3EEN7Ty>
- Timmerman, M. E., & Lorenzo-Seva, U. (2011). Dimensionality assessment of ordered polytomous items with parallel analysis. *Psychological Methods*, 16(2), 209. doi:10.1037/a0023353
- Udn. (2020). 2020 is the first year of Taiwan Podcast! How does the “broadcast version” YouTuber use voice to attract stans? Retrieved from <https://topic.udn.com/event/Podcast>
- Van de Schoot, R., Lugtig, P., & Hox, J. (2012). A checklist for testing measurement invariance. *European Journal of Developmental Psychology*, 1(4), 486–492. doi:10.1080/17405629.2012.686740
- Wang, J., & Wang, X. (2019). *Structural equation modeling: Applications using Mplus*. John Wiley & Sons.
- Williams, B., Onsman, A., & Brown, T. (2010). Exploratory factor analysis: A five-step guide for novices. *Australasian Journal of Paramedicine*, 8(3), 3. doi:10.33151/ajp.8.3.93
- Wimmer, R. D., & Dominick, J. R. (2009). Research in media effects. In *Mass media research: An introduction* (9th ed). Boston, MA: Cengage Learning.
- Wrather, K. (2016). Making ‘Maximum Fun’ for fans: Examining podcast listener participation online. *Radio Journal: International Studies in Broadcast & Audio Media*, 14(1), 43–63.
- Yahoo News. (2021). Digital advertising marketing review: FB, IG, LINE, Google, ASA, Podcast. Retrieved from <https://bit.ly/3sDAWnB>
- Yeh, C. C. (2017). An investigation of a podcast learning project for extensive listening. In *Asian-focused ELT research and practice: Voices from the far edge*, IDP Education (Cambodia) Ltd. (pp. 87).

## Appendix 1

### Preliminary Podcast Motivations and Usage Scale Items

---

#### Factor 1: Information-seeking/Education; 7 items

---

- 1) I can learn something new from listening to podcasts
- 2) Listening to podcasts helps me understand the latest news, trends and issues
- 3) I like to learn about other people's experiences through listening to podcasts
- 4) I can enrich myself when I listen to podcasts
  - 5) Listening to podcasts helps me acquire knowledge from different areas
  - 6) I think there is abundant information I can get when I listen to podcasts
  - 7) Compared to other media, listening to podcast can help me understand a certain topic

#### Factor 2: Entertainment; 6 items

- 8) Listening to podcasts is my entertainment
- 9) Listening to podcasts is fun
  - 10) Listening to podcasts helps me relieve stress
  - 11) Listening to podcasts is relaxing
  - 12) Listening to podcasts helps me balance my life
  - 13) Listening to podcasts makes me happy

#### Factor 3: Companionship; 5 items

- 14) Listening to podcasts helps me pass the time
  - 15) Listening to podcasts makes me feel accompanied
- 16) Listening to podcasts makes me feel less lonely
- 17) Listening to podcasts makes me feel the host is one of my friends
- 18) I like my hosts to share their personal stuff during the show

#### Factor 4: Identification/ Social interaction; 7 items

- 19) Listening to podcasts improves my understanding about what kind of person I am
- 20) Listening to podcasts helps me know more about myself
- 21) When my favorite podcasts share their views, I may agree with them.
- 22) Listening to podcasts inspires my ideas or thoughts
  - 23) I can share with others about what I have heard from the podcasts
- 24) I have searched my favorite podcasts' information online or followed their social media
  - 25) I am interested in the discussions and interactions on my favorite podcasters' social media

#### Factor 5: Convenience/Multi-tasking; 5 items

- 26) Listening to podcasts is a convenient way to get information
  - 27) Listening to podcasts is easier to get information compared to other forms of media
  - 28) I like to listen to podcasts because I can do other things while listening to it
    - 29) When I am doing other things while listening to podcasts, the time goes by faster
    - 30) Listening to podcasts helps me make good use of the fragmented or in-between time
-